explain symmetrix about x and y axis and symmetric about origin in easy way with examples

Symmetry ka simple idea: **agar aap figure ko "flip" karo aur wo bilkul same lage**, to wo symmetric hai.

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1) Symmetric about x-axis

Meaning: x-axis ke across "mirror image" hota hai.
Har point (x, y) ka mirror point $(x, -y)$ hota hai.

Test (graph/function)

Agar equation me **y ko -y** karne se equation **same** rahe → x-axis symmetry.

Examples

- $y = x^2$
  y positive hai, mirror me $-y = x^2$ (same equation nahin banti) → **x-axis symmetric nahi**.
- $x^2 + y^2 = 25$ (circle)
  y ko -y karo: $x^2 + (-y)^2 = x^2 + y^2 = 25$ → **x-axis symmetric** ✅
  (Actually circle x-axis, y-axis, origin sab ke about symmetric hota hai.)
- $y^2 = 4x$ (right-opening parabola)
  y ko -y karo: $(-y)^2 = y^2$ → equation same → **x-axis symmetric** ✅
  (Kyunkay upar aur neeche same curve banta hai.)

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2) Symmetric about y-axis

Meaning: y-axis ke across mirror image hota hai.
Har point (x, y) ka mirror point $(-x, y)$ hota hai.

Test

Equation me **x ko -x** karne se equation same rahe → y-axis symmetry.

Examples

- $y = x^2$
  x ko -x karo: $y = (-x)^2 = x^2$ → same → **y-axis symmetric** ✅
- $y = x^3$
  x ko -x karo: $y = (-x)^3 = -x^3$ (same nahi) → **y-axis symmetric nahi**.
- $x^2 + y^2 = 25$
  x ko -x karo: $(-x)^2 + y^2 = x^2 + y^2$ → same → **y-axis symmetric** ✅

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3) Symmetric about origin

Meaning: origin ke through "180° rotation" karo, figure same lage.
Har point (x, y) ka opposite point $(-x, -y)$ hota hai.

Test

Equation me **x → -x** aur **y → -y** dono karo; agar equation same rahe → origin symmetry.

Examples

- $y = x^3$
  x→-x, y→-y: $-y = (-x)^3 = -x^3$ → multiply -1: $y = x^3$ → same → **origin symmetric** ✅
- $y = x$
  $-y = -x \rightarrow y = x$ → **origin symmetric** ✅
- $y = x^2$
  $-y = (-x)^2 = x^2 \rightarrow y = -x^2$ (different) → **origin symmetric nahi**.
- $xy = 1$
  $(-x)(-y) = xy$ → same → **origin symmetric** ✅

Quick Memory Trick

- **x-axis symmetry:** sign sirf **y** ka change → $(x, y) \leftrightarrow (x, -y)$
- **y-axis symmetry:** sign sirf **x** ka change → $(x, y) \leftrightarrow (-x, y)$
- **origin symmetry:** dono signs change → $(x, y) \leftrightarrow (-x, -y)$

Agar aap 3–4 equations bhej do, main unko quickly check karke bata dunga kaun si x-axis / y-axis / origin ke about symmetric hain.